WINDOWS SETUP GUIDE

Install PostgreSQL and pgAdmin

From download to your first query, step by step

Windows · 64-bit

Beginner friendly

Free and open source



The plan: three phases

1

Prepare

Check your PC

Choose a password

Download the installer

2

Install

Click through 9 wizard steps

Keep most defaults

Set your password

3

Connect

Open pgAdmin

Create a database

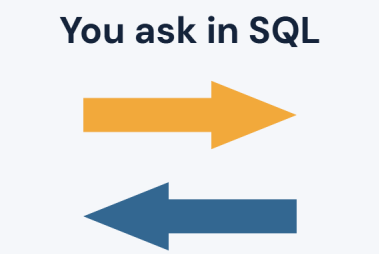
Run your first query

Everything is free. The one thing to remember: **your postgres password**.

Two programs, two jobs

pgAdmin 4

The client: the window where you click and type



PostgreSQL

The server: stores and protects your data

`They talk to each other on localhost, port 5432`

> Think of a restaurant: pgAdmin is the menu and the waiter, PostgreSQL is the kitchen.

Before you start: four checks

Windows 64-bit

Settings, System, About shows System type: 64-bit.

Admin rights

You must be able to click Yes when Windows asks for permission.

Simple user name

Use English letters and digits. Special characters can break the installer.

A password ready

Choose the postgres password now and write it down.

Tip: pick a simple password you will not forget. Never reuse a real password from email or banking.

Download the installer

`postgresql.org/download/windows`

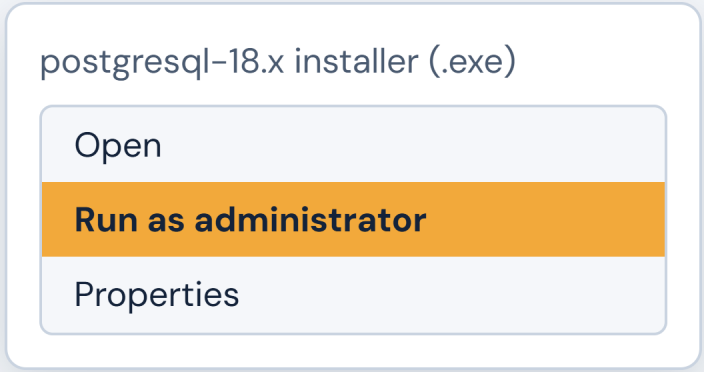
1. Open the address above in your browser.
2. Click **Download the installer** (interactive installer by EDB).
3. Choose the newest version, **18.x**, for **Windows x86-64**.
4. Click **Download** and wait for the .exe file to finish.

You should now have this file in Downloads: `postgresql-18.x-1-windows-x64.exe`

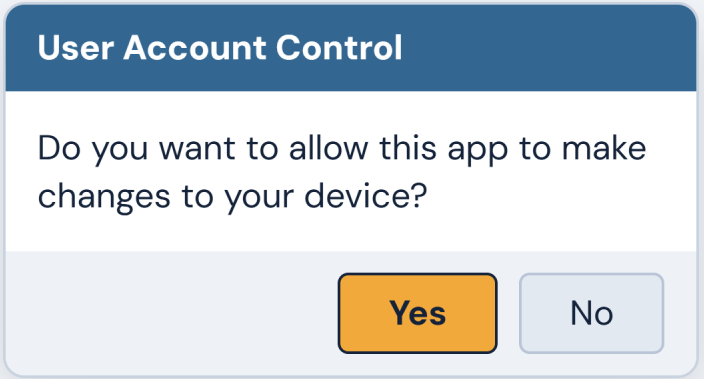
STEP 1 OF 9

Start the installer

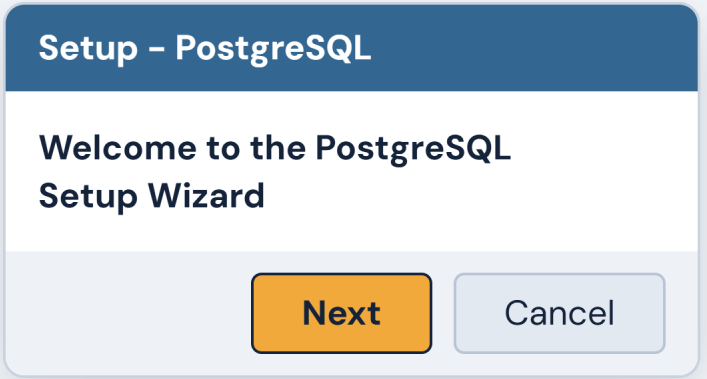
1 Right-click the file



2 Click Yes



3 Welcome: click Next



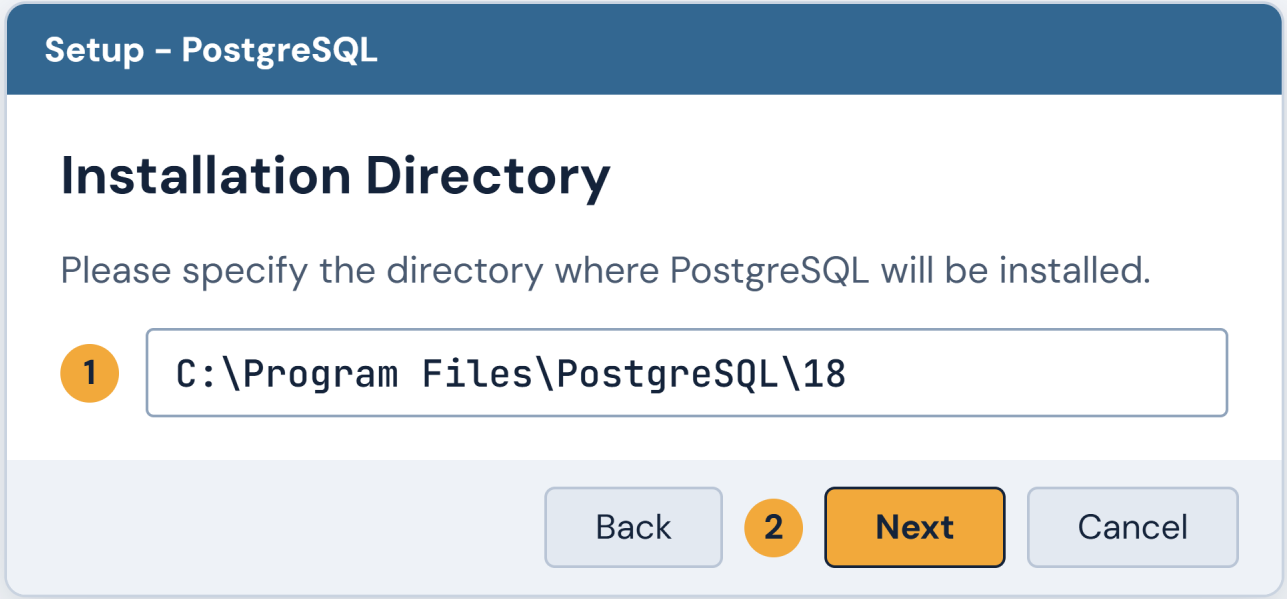
Tip: if Run as administrator is missing, double-click the file and give the admin password when asked.

STEP 2 OF 9

Choose the install folder

1. Keep the default folder. Do not change it.
2. Click Next.

> **Tip:** the number at the end of the path (18) is the PostgreSQL version.

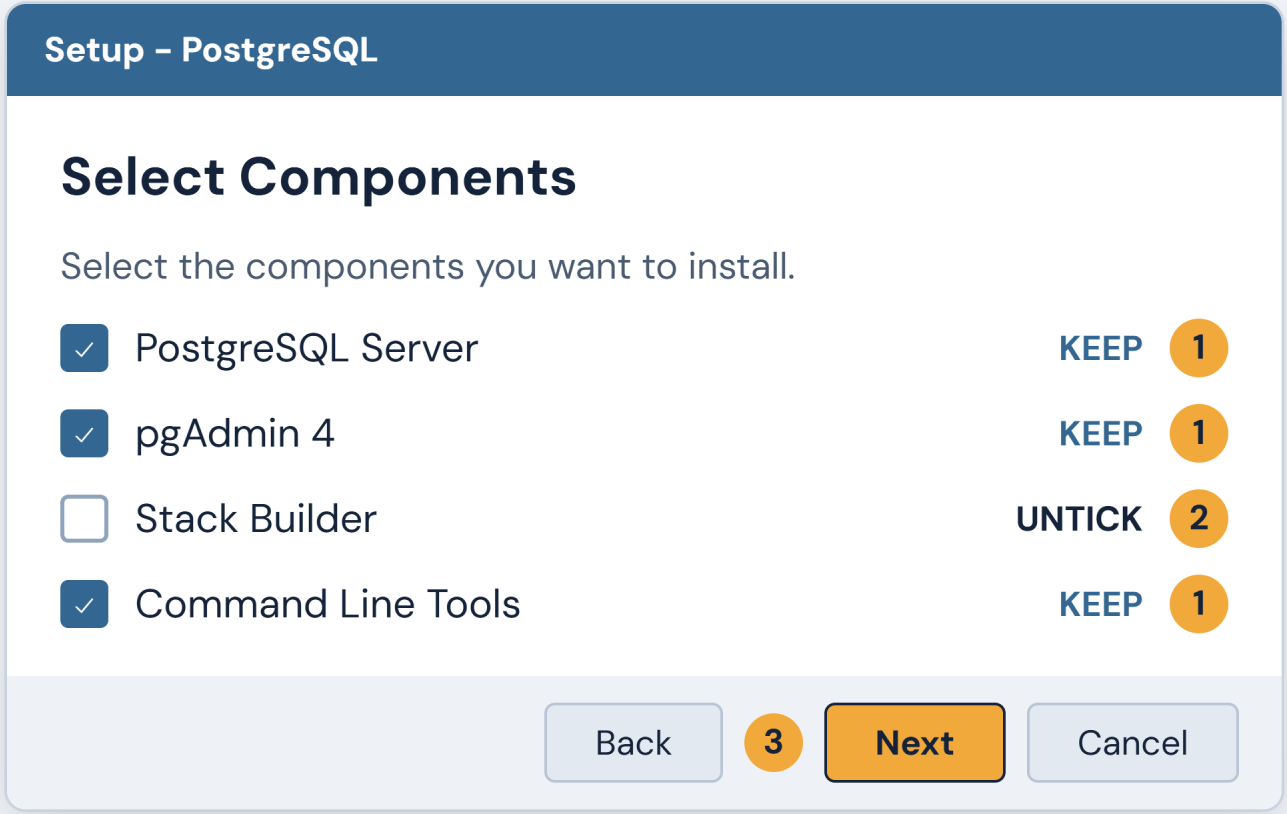


STEP 3 OF 9

Select the components

1. Keep three boxes ticked: Server, pgAdmin 4, Command Line Tools.
2. Untick Stack Builder. You do not need it.
3. Click Next.

Important: pgAdmin 4 must stay ticked. It is the tool we use next.

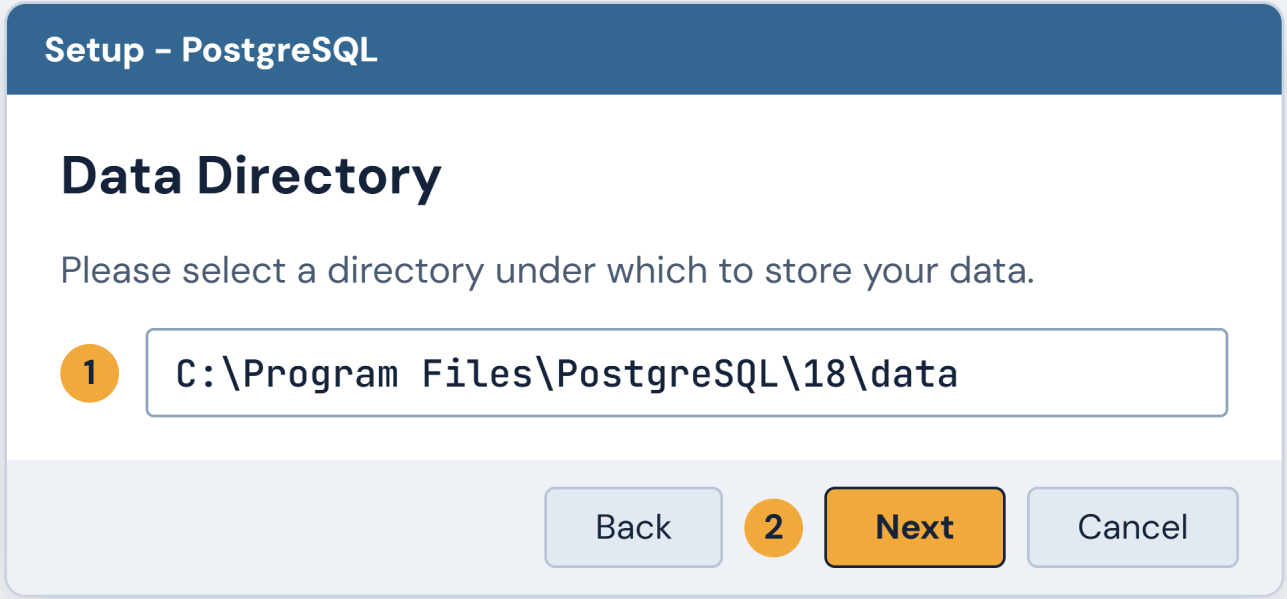


STEP 4 OF 9

Choose the data folder

1. Keep the default data folder.
2. Click Next.

Tip: this is where your databases are stored on disk. Do not delete this folder.

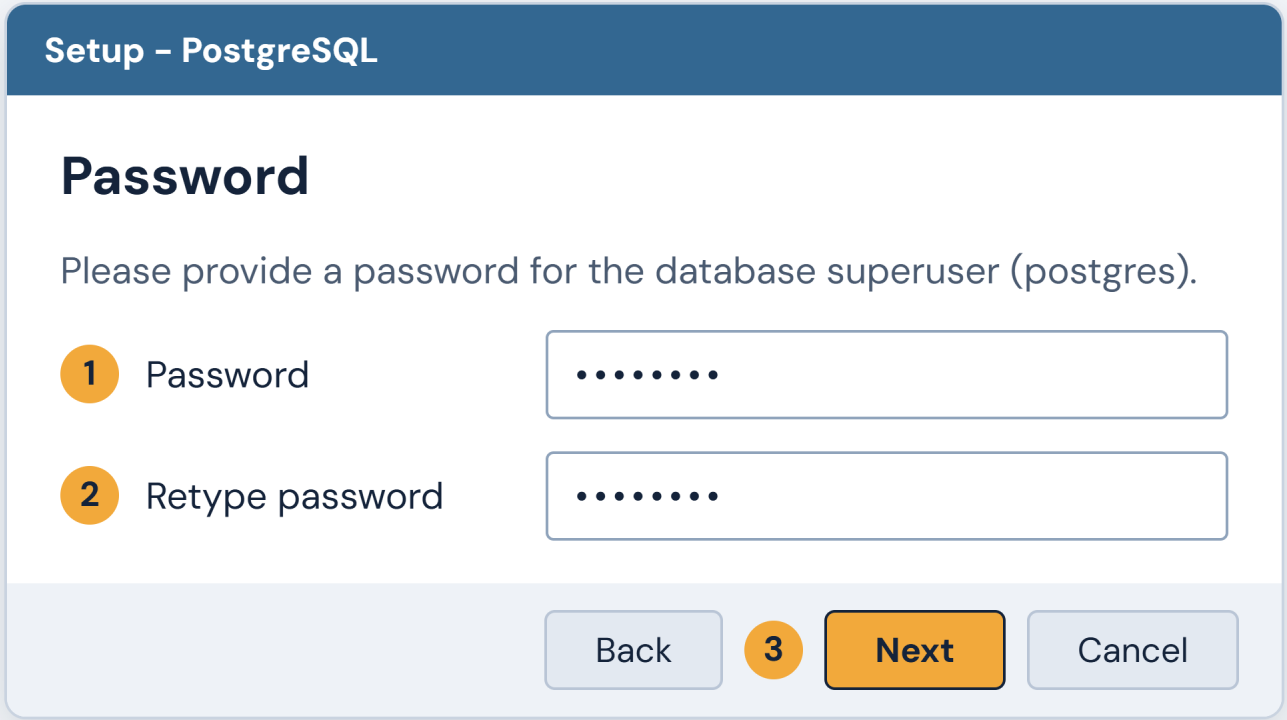


STEP 5 OF 9

Set the postgres password

1. Type the password you chose.
2. Type the exact same password again.
3. Click Next.

Write it down now. The user name is always **postgres**. You need both in pgAdmin.

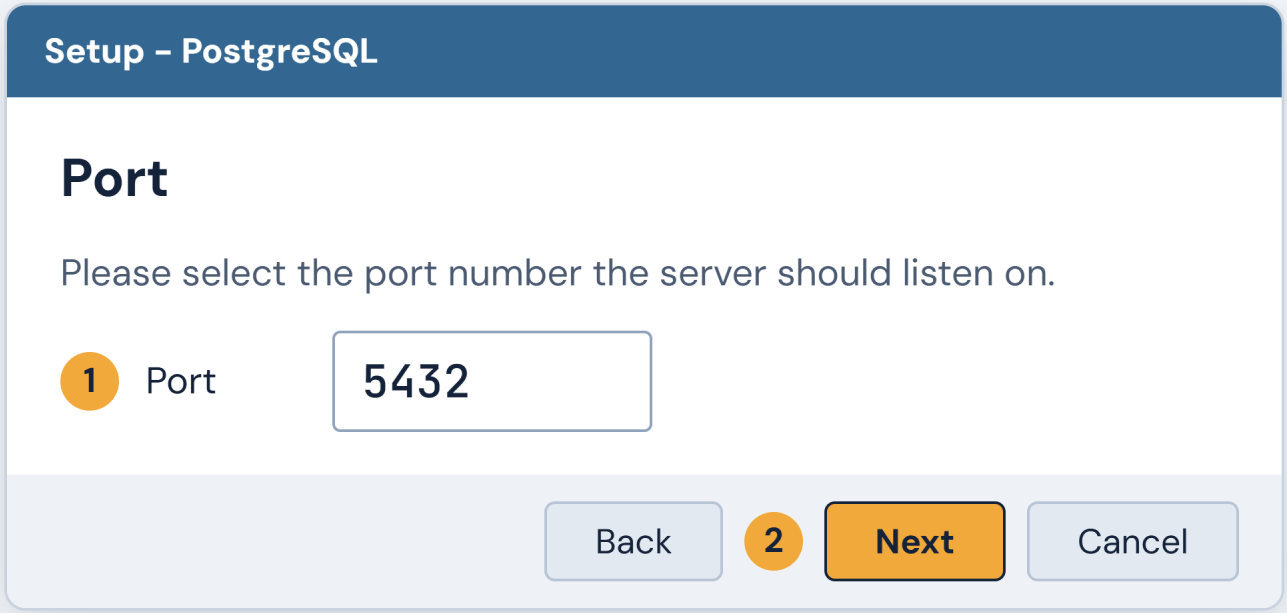


STEP 6 OF 9

Keep the port

1. Keep the default port: 5432.
2. Click Next.

> **Tip:** if another program already uses 5432, type 5433 instead and remember it for pgAdmin.

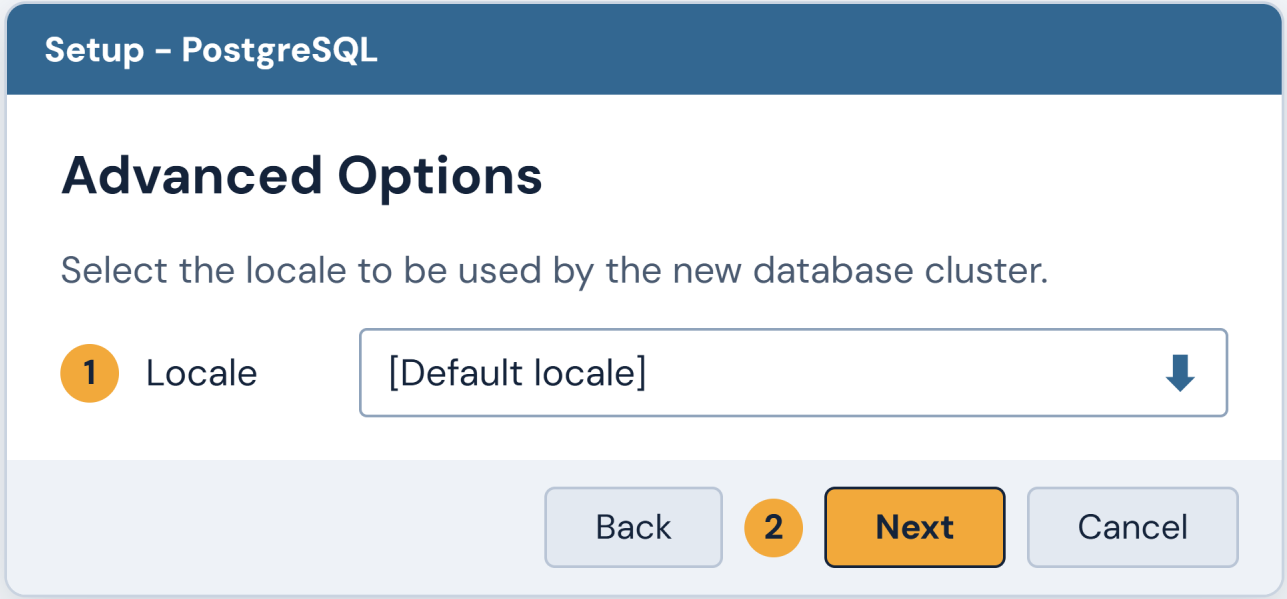


STEP 7 OF 9

Keep the default locale

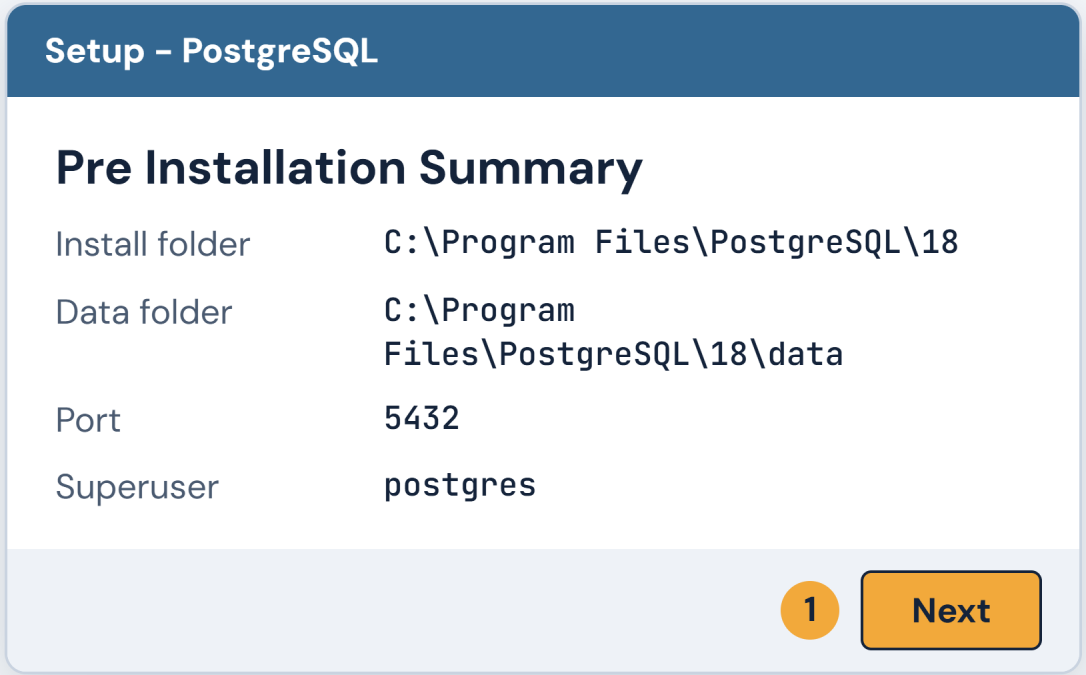
1. Leave Locale on [Default locale].
2. Click Next.

> **Tip:** the default locale is simply your Windows language and region settings.

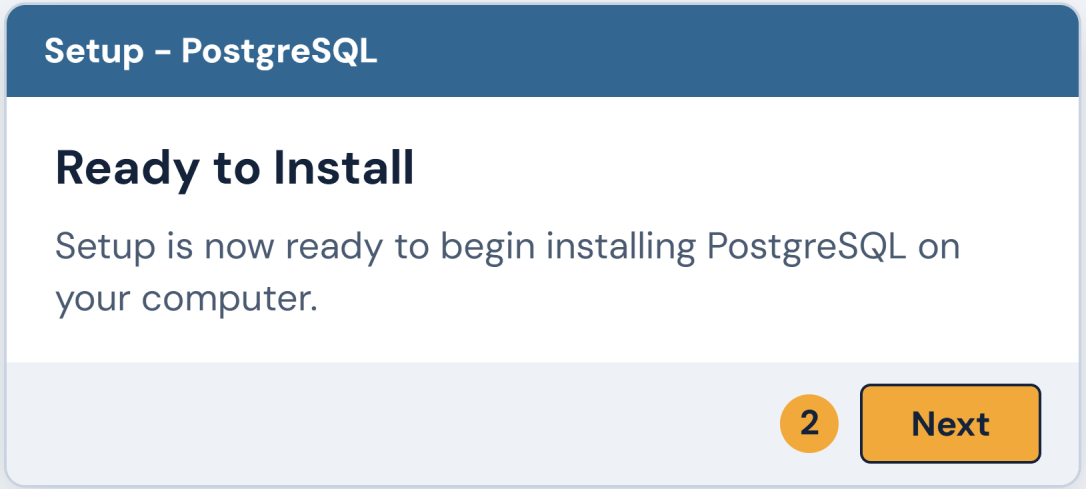


STEP 8 OF 9

Review and start



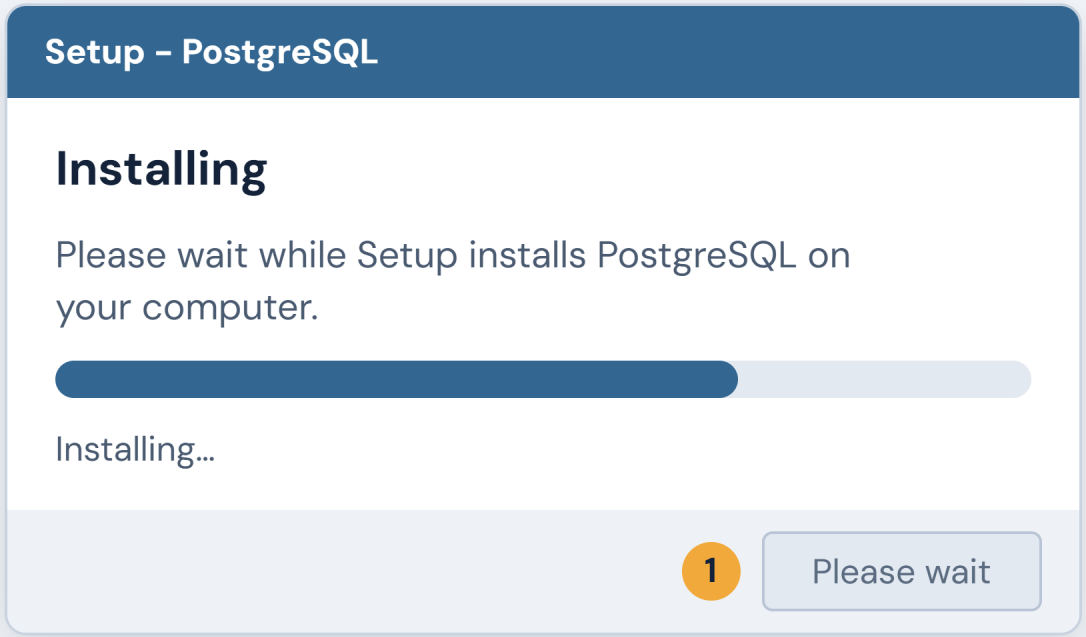
1 Check the folders, the port 5432 and the user name, then click Next.

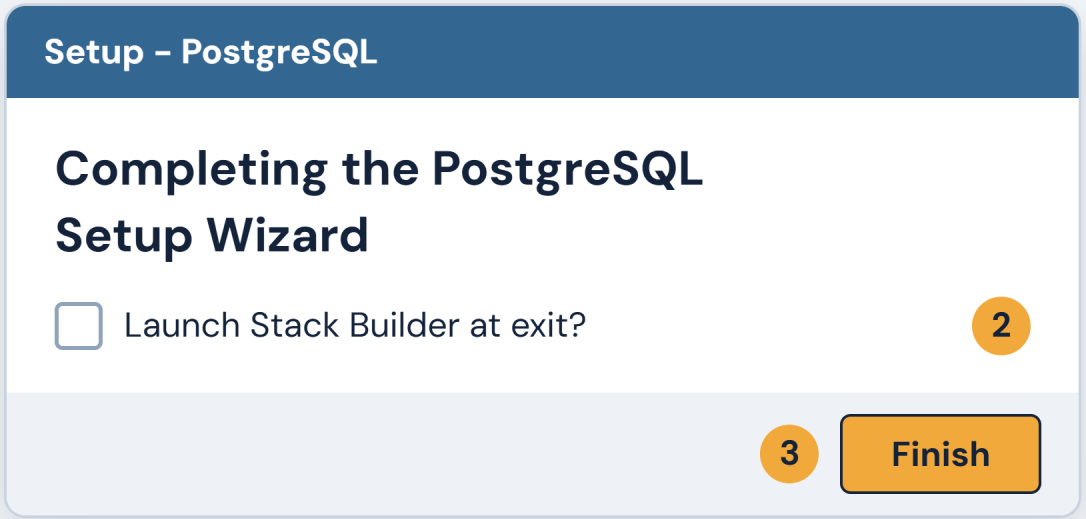


2 Click Next once more. The installation starts.

STEP 9 OF 9

Wait, then finish





1. Wait while the progress bars fill. Do not close the window.
2. Untick the Stack Builder box.
3. Click Finish. PostgreSQL is installed.

AFTER 1 OF 6

Check that the server runs

1. Press the Windows key, type Services, press Enter.
2. Find postgresql-x64-18 in the list.
3. Status says Running? Good. If not, right-click it and choose Start.

> **Tip:** the server now starts automatically whenever Windows starts.

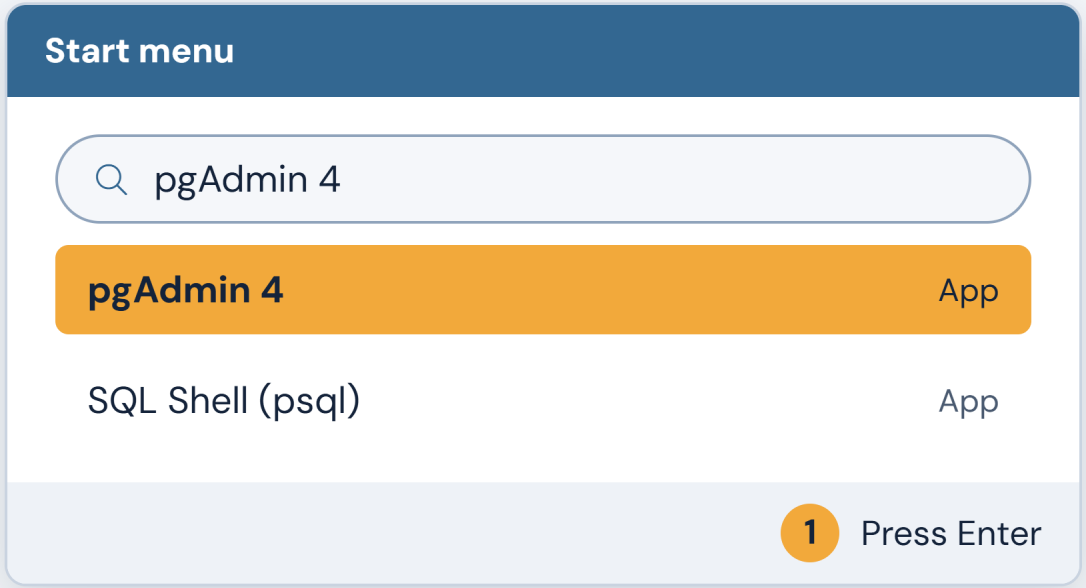
Services (Local)

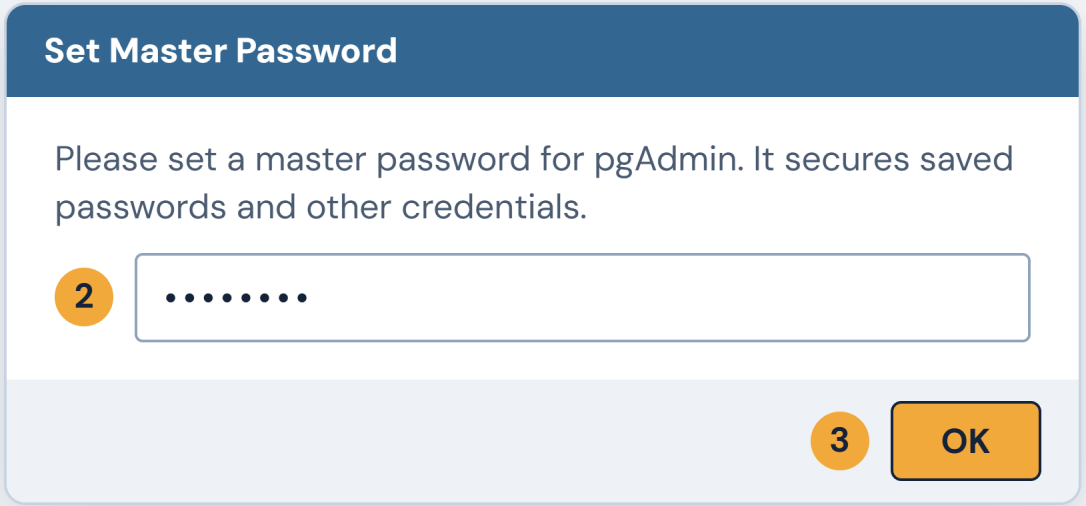
Name	Status	Startup Type
Print Spooler	Running	Automatic
postgresql-x64-18 - PostgreSQL Server 18	Running	Automatic
Windows Audio	Running	Automatic

2 3 Highlighted row: the PostgreSQL server

AFTER 2 OF 6

Open pgAdmin 4



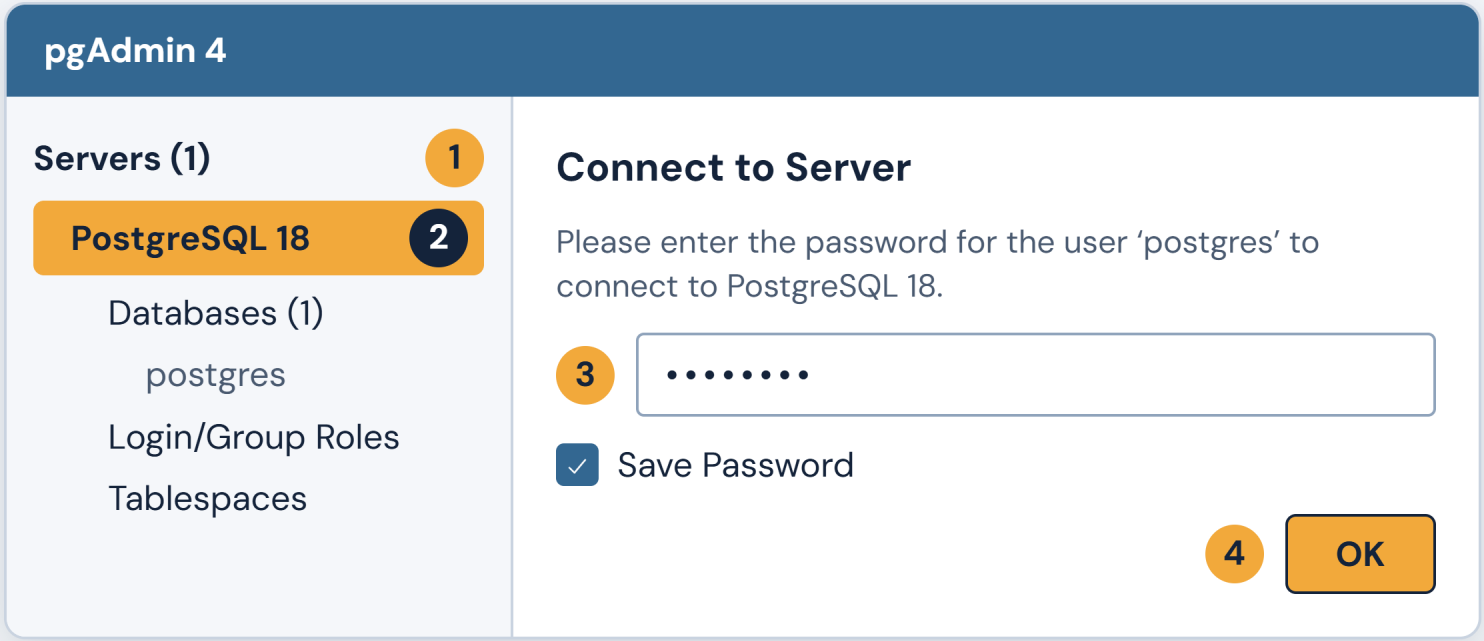


1. Click Start, type pgAdmin 4, press Enter. The first launch can take a moment.
2. Set a master password if asked. It is not the postgres password.
3. Click OK. The pgAdmin window opens.

AFTER 3 OF 6

Connect to the server

1. Expand Servers on the left.
2. Double-click PostgreSQL 18.
3. Type the postgres password from the installer.
4. Tick Save Password, then click OK.

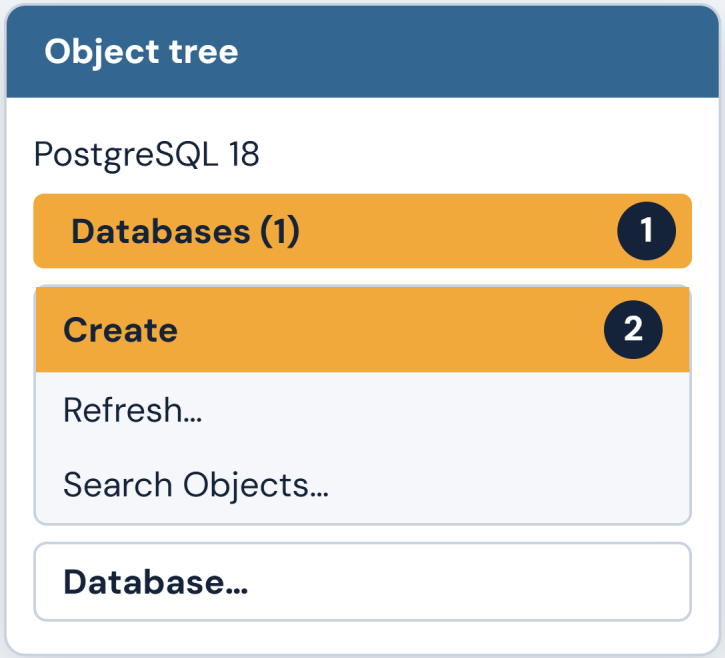


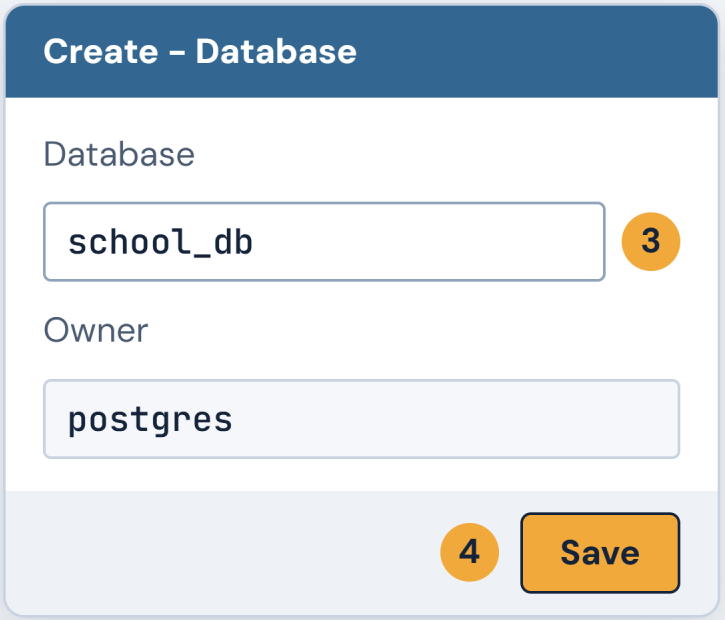
AFTER 4 OF 6

Create your first database

1. Right-click Databases in the tree.
2. Choose Create, then Database...
3. Type a name, for example school_db.
4. Click Save.

Tip: use lowercase letters and underscores in names.

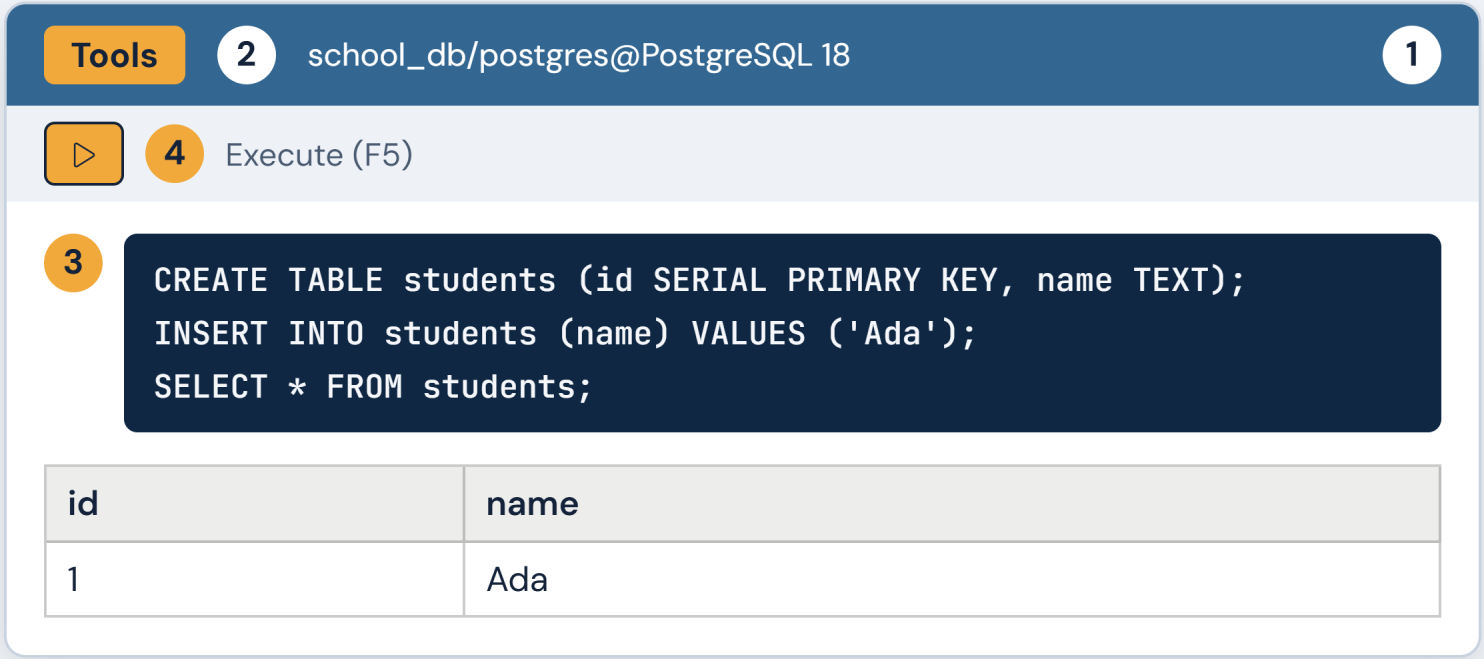




AFTER 5 OF 6

Run your first query

1. Click school_db in the tree.
2. Open Tools, then Query Tool.
3. Type the three lines of SQL.
4. Click the Play button or press F5. Results appear below.



AFTER 6 OF 6 · OPTIONAL

Use psql in a terminal

SQL Shell (psql)

```
Server [localhost]:                  Enter
Database [postgres]:                 Enter
Port [5432]:                         Enter
Username [postgres]:                 Enter
Password for user postgres:          Type it
postgres=#
```

“psql is not recognized”?

Add the bin folder to the PATH:

```
C:\Program Files\PostgreSQL\18\bin
```

1. Run SystemPropertiesAdvanced.exe
2. Environment Variables, then System variables, Path, Edit
3. New, paste the folder above, OK
4. Open a new Command Prompt and run:

```
psql -U postgres
```

Something went wrong? Six quick fixes

Port 5432 is in use

Run the installer again and type 5433 on the Port screen. Use 5433 in pgAdmin too.

psql is not recognized

Add the PostgreSQL bin folder to the PATH (see the previous slide), then open a new Command Prompt.

Forgot the password

Ask your teacher. With no data to lose, uninstall and install again with a new password.

Installer shows an error

Use English letters and digits in your Windows user name, and run the installer as administrator.

pgAdmin cannot connect

Check three things: the service says Running, the password is right, the port is 5432.

Start over

Settings, Apps, PostgreSQL 18, Uninstall. Then run the installer again.

You did it: recap and cheat sheet

- Downloaded the installer from postgresql.org
- Installed the server, pgAdmin 4 and command line tools
- Checked that the service is Running
- Connected with pgAdmin and created school_db
- Ran your first SQL query

YOUR CONNECTION DETAILS

Host	`localhost`
Port	`5432`
User	`postgres`
Password	the one you wrote down
Database	`school_db`